1 Time taken = 1.5 hours (or 1 hour 30 minutes) **[B1]**

$120 \div 1.5 = 80$ **[B1]**

[Total Marks: 2]

2 Time = 1.5 hours **[B1]**

Average Speed = $\frac{120}{1.5}$ **[M1]**

80 **[A1]**

[Total Marks: 3]

3 Time = $4\frac{25}{60}$ hours or 4.416... hours **[M1]**

Average Speed = $\frac{385}{4.416...}$ **[M1]**

87.2 (correct to 1 decimal place) **[A1]**

[Total Marks: 3]

4 (a) 45×1000 or $\frac{45}{60}$ **[M1]**

$\frac{45000}{60} = 750$ **[A1]**

(b) Base area = $150 \times 80 = 12000$ cm^2 **[M1]**

Rate of height increase = $\frac{\text{Volume per second}}{\text{Base area}} = \frac{750}{12000}$ **[M1]**

0.0625 **[A1]**

[Total Marks: 5]

5 (a) $\frac{350}{100} \times 6.4$ **[M1]**

22.4 **[A1]**

(b) Cost in euros = $22.4 \times 1.35 = 30.24$ **[M1]**

Cost in dollars = $30.24 \div 0.85$ **[M1]**

35.58 **[A1]**

[Total Marks: 5]

6 (a) Volume in $m^3 = 1.8 \times 1.5 \times 1.2 = 3.24$ **[M1]**

$3.24 \times 1000 = 3240$ litres **[A1]**

(b) Time in seconds = $\frac{3240}{2.5} = 1296$ **[M1]**

Time in minutes $= \dfrac{1296}{60} = 21.6$ **[A1]**

(c) Rate $= \dfrac{\text{Volume}}{\text{Time}} = \dfrac{3240}{45}$ **[M1]**

$x = 72$ **[A1]**

[Total Marks: 6]

7 (a) Time A to B $= \dfrac{d}{v}$ **[M1]**

Time B to C $= \dfrac{2d}{v+15}$ **[M1]**

Total Time $T = \dfrac{d}{v} + \dfrac{2d}{v+15} = \dfrac{d(v+15)+2dv}{v(v+15)} = \dfrac{dv+15d+2dv}{v(v+15)} = \dfrac{3dv+15d}{v(v+15)}$ **[A1]**

(b) Substitute $d = 120$ and $T = 4.5$ into the equation from part (a): $4.5 = \dfrac{3(120)v + 15(120)}{v(v+15)}$ **[M1]**

$4.5(v^2 + 15v) = 360v + 1800$ **[M1]**

$4.5v^2 + 67.5v - 360v - 1800 = 0$ **[M1]**

$4.5v^2 - 292.5v - 1800 = 0 \implies divideby1.5toget3\text{v}^2 - 195v - 1200 = 0$ **[A1]**

[Total Marks: 7]

8 (a) $\dfrac{30}{x} + \dfrac{18}{x-4} = 2.5$ **[M1]**

$\dfrac{30(x-4)+18x}{x(x-4)} = 2.5$ **[M1]**

$48x - 120 = 2.5(x^2 - 4x)$ **[M1]**

$48x - 120 = 2.5x^2 - 10x \implies 2.5x^2 - 58x + 120 = 0 \implies 5x^2 - 116x + 240 = 0$ **[A1]**

(b) $x = \dfrac{-(-116) \pm \sqrt{(-116)^2 - 4(5)(240)}}{2(5)}$ **[M1]**

$x = \dfrac{116 \pm \sqrt{13456 - 4800}}{10} = \dfrac{116 \pm \sqrt{8656}}{10}$ **[M1]**

$x = 20.90$ or $x = 2.30$ **[A1]**

[Total Marks: 7]

9 (a) $\dfrac{15}{x} + \dfrac{12}{x-4} = 3.5$ **[M1]**

Multiply by $2x(x-4)$ to clear denominators (or equivalent): $30(x-4) + 24x = 7x(x-4)$ **[M1]**

$54x - 120 = 7x^2 - 28x \implies 7x^2 - 82x + 120 = 0$ **[A1]**

(b) (i) $(7x - 12)(x - 10) = 0$ OR $\dfrac{82 \pm \sqrt{(-82)^2 - 4(7)(120)}}{14}$ **[M2]**

If M0, allow M1 for $(7x \pm a)(x \pm b) = 0$ where $ab = 120$ or $7b \pm a = -82$

$x = 10$, $x = \dfrac{12}{7}$ (or 1.71) **[A1]**

(ii) Identifies valid speed $x = 10$ (since $x - 4 > 0 \implies x > 4$) **[B1]**

Time $= \dfrac{15}{10} = 1.5$ **[B1]**

[Total Marks: 8]

10 (a) Time $= 2.25$ or $\dfrac{9}{4}$ hours **[M1]**

$180 \div 2.25 = 80$ **[A1]**

(b) Time in seconds $= 450 \div 1.5 = 300$ **[M1]**

$300 \div 60 = 5$ **[A1]**

[Total Marks: 4]

11 (a) Fuel used $= \dfrac{325}{100} \times 8.4$ or 27.3 **[M1]**

27.3×1.35 **[M1]**

36.855 or 36.86 **[A1]**

(b) 24 minutes $= 0.4$ hours or $\dfrac{2}{5}$ hours **[B1]**

Distance $= 75 \times 4.4$ **[M1]**

330 **[A1]**

[Total Marks: 6]

12 (a) $\dfrac{2400}{v}$ **[B1]**

(b) $\dfrac{2400}{v} + \dfrac{2400}{v+40} = 10.5$ **[M1]**

$2400(v + 40) + 2400v = 10.5v(v + 40)$ **[M1]**

$2400v + 96000 + 2400v = 10.5v^2 + 420v$ **[M1]**

$10.5v^2 - 4380v - 96000 = 0$ **[A1]**

Multiply by $\dfrac{2}{3}$ to get $7v^2 - 2920v - 192000 = 0$ **[A1]**

(c) $v = \dfrac{-(-2920) \pm \sqrt{(-2920)^2 - 4(7)(-192000)}}{2(7)}$ **[M1]**

$v = 475.25...$ or $v = -57.68...$ (discard negative) **[A1]**

475 (correct to nearest integer) **[A1]**

[Total Marks: 7]